Priority-Aware Firm and Flexible Capacity Quantification for Large-Load Interconnection

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***Abstract*—We study how to unlock existing transmission capacity to accelerate large-load interconnection under differentiated quality-of-service (QoS) requirements. We establish an efficient framework to quantify firm and flexible hosting capacities and assess marginal values of grid upgrades. Specifically, we define multiple flexible capacity tiers with differentiated QoS limits and design a two-layer priority rule to allocate curtailment among tiers. We then formulate a mixed-integer chance-constrained model to jointly quantify firm and flexible capacities under QoS and priority constraints. To improve computational efficiency, we derive an equivalent aggregate formulation that eliminates individual curtailment variables and priority binaries without changing the feasible capacity set. Numerical results show the computational advantage of aggregation. Multi-tier flexible capacity increases total admitted capacity by 45.0% over the firm-only hosting capacity.**

***Index Terms*—Large-load interconnection, flexible capacity, curtailment priority.**

I. Introduction

The rapid growth of large electricity loads, including AI data centers, is placing increasing pressure on transmission grid interconnection [1]. However, transmission grid expansion is slow and capital-intensive, making it difficult to keep pace with large-load interconnection requests. This paper focuses on accommodating this demand surge before network upgrades while meeting differentiated quality-of-service (QoS) requirements. Specifically, we consider multi-tier flexible interconnection, where different tiers have distinct QoS limits. Unlike firm capacity, which must be served under all planning conditions, flexible capacity (also known as non-firm load service) allows curtailment subject to agreed QoS limits.

Existing studies demonstrate the benefits of such flexibility for load growth in U.S. power systems [2], distribution-level interconnection [3], and data-center siting in transmission networks [4]. However, large-load customers may differ in their QoS requirements, both in terms of allowed curtailment depth and the probability of exceeding that depth. These studies do not explicitly consider multiple flexible capacity tiers with differentiated QoS requirements.

Another challenge is how to allocate curtailment when multiple flexible tiers share a bus. Their QoS limits do not specify how curtailment should be distributed among them. Priority-based allocation provides one efficient approach and has been studied in electricity markets [5]. PJM has also proposed curtailment priorities for certain large-load services [6]. However, these studies do not determine how much firm and tier-specific flexible capacity a transmission network can jointly admit while satisfying network and QoS constraints.

Our contributions addressing these gaps are twofold. First, we develop a two-layer curtailment priority rule and formulate a unified mixed-integer chance-constrained model that maximizes the preference-weighted firm and flexible interconnection capacity subject to network, priority, and QoS limits. We then establish an exact aggregate reformulation that eliminates individual curtailment variables and priority binaries without changing the feasible set of capacity decisions. Second, to improve computational efficiency and guide grid upgrades, we first approximate the aggregate model by a linear program (LP) whose feasible solutions provably satisfy the QoS limits, then derive marginal upgrade values from the dual variables that indicate how much additional hosting capacity can be unlocked by the upgrade in transmission or generation capacity. Numerical results on the PJM 5-bus system show that aggregation reduces solution time from 5801.5 s to 1.56 s while preserving the optimal objective value. On the IEEE RTS 73-bus system, results show that multi-tier flexible capacity increases total admitted capacity by 45.0% over the firm-only hosting capacity.

The paper is organized as follows. Section II defines the capacity products and formulates the individual model under the two-layer curtailment priority rule. Section III derives the equivalent aggregate model and the LP approximation, whose duals give marginal upgrade values. The experiments are given in Section IV.

II. Individual Capacity Quantification

In this section, we first introduce the QoS limits and the curtailment priority rule, and then formulate an individual model for capacity quantification. The grid planner performs this quantification to determine firm and flexible interconnection capacities. In the next section, we derive an equivalent aggregate model and the LP approximation to improve computational efficiency and obtain marginal upgrade information.

A. Firm and Flexible Capacity Products

Let $\mathcal{N}$ and $\mathcal{J}$ denote the sets of buses and transmission lines, respectively, and let $\mathcal{I} \subseteq \mathcal{N}$ denote the candidate interconnection buses. The planning horizon is $\mathcal{T} := \{1, \dots, T\}$, and $\mathcal{K} := \{1, \dots, K\}$ indexes flexible service tiers with

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different QoS requirements. Existing nodal demand is represented by a random trajectory $\mathbf{D} := (\mathbf{D}_t)_{t\in\mathcal{T}}$, where $\mathbf{D}_t := (D_{it})_{i\in\mathcal{N}} \in \mathbb{R}^{|\mathcal{N}|}$. Throughout the paper, $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the distribution of $\mathbf{D}$. Unless otherwise specified, $i \in \mathcal{I}$, $k \in \mathcal{K}$, and $t \in \mathcal{T}$ throughout the formulations. Bold symbols denote vectors and matrices, and vector inequalities are componentwise. Unindexed bold decision symbols collect the corresponding variables over all relevant indices.

At bus i, *firm capacity* $c_i^{\mathrm{F}} \geq 0$ is served in full in every period and almost every realization. *Flexible capacity* $c_{ik} \geq 0$ belongs to service tier k and may be curtailed under specified conditions. Let $\mathbf{c}_i := (c_{ik})_{k\in\mathcal{K}}$ and $\mathbf{c}_k := (c_{ik})_{i\in\mathcal{N}}$ collect flexible capacities across tiers and buses, respectively, and let $\mathbf{c} := (c_{ik})_{i\in\mathcal{I},k\in\mathcal{K}}$ collect all flexible capacity awards. The firm-capacity vector is $\mathbf{c}^{\mathrm{F}} := (c_i^{\mathrm{F}})_{i\in\mathcal{N}}$. Each admitted flexible product requests its full capacity in every period. Product (i,k) refers to service tier k at bus i. Let y_{ikt} denote the curtailed amount of product (i,k) in period t. The delivered flexible load is then $c_{ik} - y_{ikt}$, where $0 \leq y_{ikt} \leq c_{ik}$. Define $\mathbf{y}_{kt} := (y_{ikt})_{i\in\mathcal{N}}$ for the tier-k curtailment vector. New-load capacities, request limits, and associated curtailments are extended by zero to buses outside $\mathcal{I}$.

Each QoS tier k is defined by two parameters (δ_k, ϵ_k). The parameter δ_k determines the allowed curtailment depth $\delta_k c_{ik}$, while ϵ_k limits the probability that this depth is exceeded at least once during the planning horizon:

$$\mathbb{P}\left(\max_{t\in\mathcal{T}} y_{ikt} > \delta_k c_{ik}\right) \leq \epsilon_k. \tag{1}$$

Curtailment within the allowed depth is not frequency-limited. These QoS parameters are specified in the interconnection agreement, allowing customers to plan on-site backup resources or load flexibility based on the agreed curtailment depth, while limiting the probability of deeper curtailment. Together, these provisions enable earlier interconnection under differentiated QoS requirements.

The tiers are ordered from most to least reliable, with

$$\begin{aligned} 0 \leq \delta_1 \leq \cdots \leq \delta_K < 1, \\ 0 < \epsilon_1 \leq \cdots \leq \epsilon_K < 1. \end{aligned} \tag{2}$$

Thus, a more reliable tier has no larger allowed curtailment depth or exceedance-probability limit. We use common parameters (δ_k, ϵ_k) across buses for notational simplicity; the results also extend to bus-specific reliability parameters satisfying (2) at each bus. A flexible product (i,k) is called active if $c_{ik} > 0$ and inactive otherwise.

B. The Two-Layer Curtailment Priority Rule

Equation (1) specifies the QoS requirement of each flexible product but does not determine how curtailment is allocated. We therefore adopt a priority rule for products sharing a bus. Specifically, we adopt a two-layer rule: the allowed curtailment depths of all flexible products are used before any product is curtailed beyond its allowed depth. The first layer is the within-depth layer, with tranche sizes $a_{ik} := \delta_k c_{ik}$; the second layer is the exceedance layer, with tranche sizes $b_{ik} := (1-\delta_k)c_{ik}$. Within each layer, less reliable tiers are curtailed before more reliable ones. Next, a small example of two products shows the curtailment priority. Fig. 1 illustrates this order for two products, with the bus index omitted. Curtailment first uses the 5 MW within-depth tranche of product 2, then the 2 MW within-depth tranche of product 1. Only after both within-depth tranches are exhausted does it use the 5 MW exceedance tranche of product 2, followed by the 8 MW of product 1.

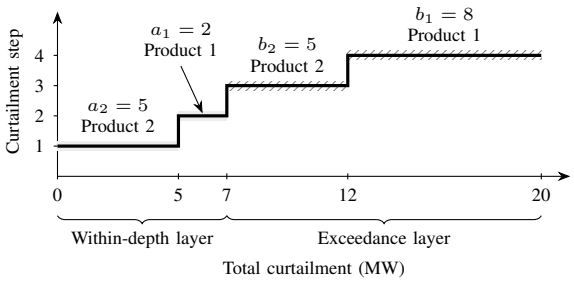

Fig. 1. A toy example for two-layer curtailment priority with $c_1 = c_2 = 10$ MW, $\delta_1 = 0.2$, and $\delta_2 = 0.5$, with the bus index omitted. Product 2 is less reliable. Curtailment follows the order $a_2 \to a_1 \to b_2 \to b_1$. Tranche widths give the amount curtailed in each step.

Let $\mathcal{M} := \{1, \ldots, 2K\}$ index all curtailment tranches at a bus. Following the two-layer rule, these tranches are curtailed in the order

$$(d_{i1}(\mathbf{c}_i), \ldots, d_{i,2K}(\mathbf{c}_i)) := (a_{iK}, \ldots, a_{i1}, b_{iK}, \ldots, b_{i1}). \tag{3}$$

Here, $d_{im}(\mathbf{c}_i)$ denotes the size of tranche m, which can be expressed as

$$d_{im}(\mathbf{c}_i) := \begin{cases} \delta_{K-m+1} c_{i,K-m+1}, & 1 \leq m \leq K, \\ (1-\delta_{2K-m+1}) c_{i,2K-m+1}, & K < m \leq 2K. \end{cases}$$

Each tranche size is linear in the flexible capacity decisions. Under this ordering, tranche $m+1$ can receive curtailment only after tranche m has been fully curtailed.

We next formulate the two-layer priority rule using linear constraints and binary variables. Let x_{imt} denote the curtailed amount assigned to tranche m in period t. To enforce the curtailment order, we introduce binary variables β_{imt} for $m = 0, \ldots, 2K$, with fixed endpoints $\beta_{i0t} = 1$ and $\beta_{i,2K,t} = 0$. For $0 < m < 2K$, setting $\beta_{imt} = 1$ indicates that tranche m is fully curtailed, allowing the next tranche to receive curtailment. The priority rule is then represented by the following constraints:

$$0 \leq x_{imt} \leq d_{im}(\mathbf{c}_i), \qquad \forall i, m, t, \tag{4a}$$

$$x_{imt} \leq M\beta_{i,m-1,t}, \qquad \forall i, m, t, \tag{4b}$$

$$x_{imt} \geq d_{im}(\mathbf{c}_i) - M(1-\beta_{imt}), \qquad \forall i, m, t, \tag{4c}$$

$$\beta_{i,m-1,t} \geq \beta_{imt}, \qquad \forall i, m, t, \tag{4d}$$

$$y_{ikt} = x_{i,K-k+1,t} + x_{i,2K-k+1,t}, \qquad \forall i, k, t, \tag{4e}$$

$$\beta_{i,0,t} = 1, \quad \beta_{i,2K,t} = 0, \qquad \forall i, t, \tag{4f}$$

$$\beta_{imt} \in \{0,1\}, \qquad \forall i, m, t. \tag{4g}$$

Let $\bar{C}_i^{\mathrm{F}}$ and $\bar{C}_{ik}$ denote the maximum firm and tier-k capacity requests at bus i, respectively, and set $\bar{\mathbf{c}}_i := (\bar{C}_{i1}, \ldots, \bar{C}_{iK})$. A valid big-M constant is $M := \max_{i,m} d_{im}(\bar{\mathbf{c}}_i)$, which bounds every feasible tranche size. Constraints (4a)–(4d), together with the fixed endpoints, ensure that a tranche receives curtailment only after all earlier

tranches are fully curtailed, leaving at most one partially curtailed tranche. Equation (4e) recovers each product's curtailment by summing its two tranche curtailments, ensuring $0 \le y_{ikt} \le c_{ik}$.

C. Mixed-Integer Chance-Constrained Quantification Model

Combining the curtailment priority rule with the generation and transmission network constraints, we formulate the following model to determine the firm and flexible capacities. The capacities $\mathbf{c}^{\mathrm{F}}$ and $\mathbf{c}$ are first-stage decisions made before $\mathbf{D}$ is realized; $\mathbf{g}$, $\mathbf{q}$, $\mathbf{y}$, $\mathbf{x}$, and $\boldsymbol{\beta}$ are recourse decision rules depending on $\mathbf{D}$. We suppress this dependence and require the operational and priority constraints to hold almost surely.

$$(\mathcal{P}_{\mathrm{ind}}) \quad \max_{\substack{\mathbf{c}^{\mathrm{F}},\mathbf{c},\mathbf{g},\mathbf{q},\\ \mathbf{y},\mathbf{x},\boldsymbol{\beta}}} W(\mathbf{c}^{\mathrm{F}},\mathbf{c}) := (\mathbf{w}^{\mathrm{F}})^{\top}\mathbf{c}^{\mathrm{F}} + \sum_{k\in\mathcal{K}} \mathbf{w}_k^{\top}\mathbf{c}_k, \tag{5a}$$

$$\text{s.t.} \quad \mathbf{0} \le \mathbf{c}^{\mathrm{F}} \le \bar{\mathbf{C}}^{\mathrm{F}},\ \mathbf{0} \le \mathbf{c}_k \le \bar{\mathbf{C}}_k, \qquad \forall k, \tag{5b}$$

$$\mathbf{q}_t = \mathbf{g}_t - \mathbf{D}_t - \mathbf{c}^{\mathrm{F}} - \sum_{k\in\mathcal{K}}(\mathbf{c}_k - \mathbf{y}_{kt}), \qquad \forall t, \tag{5c}$$

$$\mathbf{1}^{\top}\mathbf{q}_t = 0, \qquad \forall t, \tag{5d}$$

$$-\bar{\mathbf{f}} \le \mathbf{S}\mathbf{q}_t \le \bar{\mathbf{f}}, \qquad \forall t, \tag{5e}$$

$$\underline{\mathbf{g}} \le \mathbf{g}_t \le \bar{\mathbf{g}}, \qquad \forall t, \tag{5f}$$

$$-\boldsymbol{\Delta}^{-} \le \mathbf{g}_t - \mathbf{g}_{t-1} \le \boldsymbol{\Delta}^{+}, \qquad \forall t, \tag{5g}$$

$$\mathbb{P}\Big(\max_{t\in\mathcal{T}} y_{ikt} > \delta_k c_{ik}\Big) \le \epsilon_k, \qquad \forall i,k, \tag{5h}$$

$$\text{Priority constraints (4)}, \tag{5i}$$

where $\mathbf{w}^{\mathrm{F}} := (w_i^{\mathrm{F}})_{i\in\mathcal{N}}$ and $\mathbf{w}_k := (w_{ik})_{i\in\mathcal{N}}$ collect the preference weights for the capacities, and $\mathbf{g}_t := (g_{it})_{i\in\mathcal{N}}$ and $\mathbf{q}_t := (q_{it})_{i\in\mathcal{N}}$ denote nodal generation and net injection. The objective (5a) maximizes the preference-weighted admitted capacity. Unit weights give the total admitted capacity in MW. Constraint (5b) imposes the capacity request limits. Constraints (5c)–(5g) describe the lossless DC network model. Constraint (5c) defines net injection, and (5d) enforces power balance, where $\mathbf{1}$ is the all-ones vector. Constraint (5e) limits transmission flows, where $\mathbf{S} \in \mathbb{R}^{|\mathcal{J}|\times|\mathcal{N}|}$ is the power transfer distribution factor (PTDF) matrix and $\bar{\mathbf{f}}$ denotes the line ratings. Constraints (5f) and (5g) impose the generation limits $\underline{\mathbf{g}}$ and $\bar{\mathbf{g}}$ and the ramp-down/up limits $\boldsymbol{\Delta}^{-}$ and $\boldsymbol{\Delta}^{+}$, respectively, with $\mathbf{g}_0$ denoting the given initial dispatch. Constraint (5h) imposes product-level QoS limits defined in (1), while (5i) enforces the curtailment priority.

III. Aggregate Reformulation

Model $\mathcal{P}_{\mathrm{ind}}$ captures the curtailment priority and the QoS requirements of each flexible product. However, the priority binaries and chance constraints make the model computationally challenging. The key observation is that the network constraints (5c)–(5g) depend on the flexible products only through their total capacity and total curtailment at each bus. Inspired by this observation, we first develop an aggregate model $\mathcal{P}_{\mathrm{agg}}$ and prove that it is equivalent to $\mathcal{P}_{\mathrm{ind}}$. This can improve computational efficiency, with a simulation comparison in Table I. We then apply a conservative conditional value-at-risk (CVaR) approximation to the remaining chance constraints to obtain a linear program (LP). The dual information of this LP can then be used to identify candidate upgrades.

We begin by defining the total flexible capacity and aggregate curtailment at each bus $i \in \mathcal{N}$:

$$C_i(\mathbf{c}_i) := \sum_{k\in\mathcal{K}} c_{ik}, \qquad z_{it} := \sum_{k\in\mathcal{K}} y_{ikt}. \tag{6}$$

Let $\mathbf{C}(\mathbf{c}) := (C_i(\mathbf{c}_i))_{i\in\mathcal{N}}$ and $\mathbf{z}_t := (z_{it})_{i\in\mathcal{N}}$ denote the corresponding vectors. Using (6) in (5c), together with $0 \le y_{ikt} \le c_{ik}$, gives

$$\mathbf{q}_t = \mathbf{g}_t - \mathbf{D}_t - \mathbf{c}^{\mathrm{F}} - \mathbf{C}(\mathbf{c}) + \mathbf{z}_t, \quad \forall t, \tag{7a}$$

$$\mathbf{0} \le \mathbf{z}_t \le \mathbf{C}(\mathbf{c}), \quad \forall t. \tag{7b}$$

In the aggregate model, $\mathbf{z}_t$ is a recourse decision vector.

A. Aggregate Representation of Curtailment Priority

To preserve the curtailment priority in the aggregate model, we first show that the curtailment of each product can be recovered from the total curtailment at its bus. Define

$$\Gamma_{im}(\mathbf{c}_i) := \sum_{h=1}^{m} d_{ih}(\mathbf{c}_i), \qquad m = 0,\dots,2K, \tag{8}$$

as the cumulative size of the first m tranches. Following (3), m ranges from 0 to $2K$ because the two-layer priority rule has two tranches for each of the K tiers. Thus, tranche m occupies the interval $[\Gamma_{i,m-1}, \Gamma_{im}]$ of aggregate curtailment, where $\Gamma_{i0} = 0$ and $\Gamma_{i,2K} = C_i$. From (3), the within-depth tranche a_{ik} of product k is at position $K-k+1$, while its exceedance tranche b_{ik} is at position $2K-k+1$. Their starting points are

$$\begin{aligned} A_{ik}(\mathbf{c}_i) &:= \Gamma_{i,K-k}(\mathbf{c}_i) = \sum_{r>k} \delta_r c_{ir}, \\ H_{ik}(\mathbf{c}_i) &:= \Gamma_{i,2K-k}(\mathbf{c}_i) = \sum_{r\le k} \delta_r c_{ir} + \sum_{r>k} c_{ir}. \end{aligned} \tag{9}$$

Both A_{ik} and H_{ik} are linear in $\mathbf{c}_i$. For the example in Fig. 1, $H_2 = a_2 + a_1 = 7$ MW and $H_1 = a_2 + a_1 + b_2 = 12$ MW. Thus, product 2 exceeds its allowed depth when total curtailment exceeds 7 MW, and product 1 exceeds its allowed depth when total curtailment exceeds 12 MW.

The following proposition shows that aggregate curtailment uniquely determines individual curtailments under the two-layer priority rule. We fix a bus i, a period, and a realization, and omit the period index and the argument $\mathbf{c}_i$ of d_{im}, Γ_{im}, A_{ik}, and H_{ik}. The positive part is defined as $[u]_+ := \max\{u, 0\}$.

Proposition 1 (Recovery of individual curtailment)**.** *Let the capacities satisfy* (5b)*. For every $z \in [0, C_i]$, there exists an allocation satisfying* (4) *and $\sum_k y_{ik} = z$. All such allocations have the same tranche and product curtailments, given by*

$$x_{im} := \min\big\{[z - \Gamma_{i,m-1}]_+,\, d_{im}\big\}, \quad m \in \mathcal{M}, \tag{10}$$

$$\begin{aligned} y_{ik} := f_{ik}(z;\mathbf{c}_i) := {} & \min\big\{[z - A_{ik}]_+,\, a_{ik}\big\} \\ & + \min\big\{[z - H_{ik}]_+,\, b_{ik}\big\}. \end{aligned} \tag{11}$$

The proof is given in Appendix A.

We next show that the chance constraint of each product can also be expressed using aggregate curtailment. Let $Z_i := \max_{t\in\mathcal{T}} z_{it}$ denote the maximum aggregate curtailment at bus i over the planning horizon.

Proposition 2 (Equivalent chance constraints)**.** *Let $(\mathbf{y}, \mathbf{x}, \boldsymbol{\beta})$ satisfy* (4) *in every realization, and let $z_{it} := \sum_k y_{ikt}$. Then, for every active product (i,k) with $c_{ik} > 0$ and every realization,*

$$\max_{t\in\mathcal{T}} y_{ikt} > \delta_k c_{ik} \quad \Longleftrightarrow \quad Z_i > H_{ik}(\mathbf{c}_i). \tag{12}$$

Consequently, $\mathbb{P}(\max_{t\in\mathcal{T}} y_{ikt} > \delta_k c_{ik}) = \mathbb{P}(Z_i > H_{ik}(\mathbf{c}_i))$, and the individual chance constraint (5h) *of product (i,k) holds if and only if $\mathbb{P}\big(Z_i > H_{ik}(\mathbf{c}_i)\big) \le \epsilon_k$.*

The proof is given in Appendix B.

Proposition 2 is the key step in the feasibility argument of Theorem 1 below: it allows the product-level chance constraints to be imposed directly on the aggregate curtailment.

Next, we replace the individual curtailment variables with aggregate curtailment variables as follows:

$$(\mathcal{P}_{\mathrm{agg}}) \quad \max_{\mathbf{c}^{\mathrm{F}},\mathbf{c},\mathbf{g},\mathbf{q},\mathbf{z}} \; W(\mathbf{c}^{\mathrm{F}}, \mathbf{c}) \tag{13a}$$

$$\text{s.t. (5b), (5d)–(5g), (7),} \tag{13b}$$

$$\mathbb{P}\big(Z_i > H_{ik}(\mathbf{c}_i)\big) \le \epsilon_k, \quad \forall i,k. \tag{13c}$$

Theorem 1. *Under* (2)*, $\mathcal{P}_{\mathrm{ind}}$ and $\mathcal{P}_{\mathrm{agg}}$ have the same feasible set of capacity decisions $(\mathbf{c}^{\mathrm{F}}, \mathbf{c})$ and the same optimal value.*

The proof is given in Appendix C.

The intuition behind Theorem 1 is that the network constraints depend on the flexible products only through their total capacity and curtailment at each bus. For fixed capacities, the priority rule uniquely determines the individual curtailments from aggregate curtailment, and each active product's depth-exceedance event is exactly captured by the aggregate threshold condition in (13c). These properties eliminate the individual curtailment variables and priority binaries without changing the optimal value.

B. CVaR-Based Linear Program

Although the aggregate model eliminates the priority binaries, its chance constraints (13c) remain nonconvex. For each product (i,k), define the loss $L_{ik} := Z_i - H_{ik}(\mathbf{c}_i)$, so that (13c) becomes $\mathbb{P}(L_{ik} > 0) \le \epsilon_k$. We conservatively replace this chance constraint with $\mathrm{CVaR}_{1-\epsilon_k}(L_{ik}) \le 0$, where $\mathrm{CVaR}_{1-\epsilon}(L)$ denotes the upper-tail conditional value-at-risk (CVaR) at confidence level $1-\epsilon$.

To obtain a finite LP, we consider S demand trajectories $\mathbf{D}^s := (\mathbf{D}^s_t)_{t\in\mathcal{T}}$, indexed by $s \in \mathcal{S} := \{1,\dots,S\}$, with probabilities $p_s > 0$ and $\sum_{s\in\mathcal{S}} p_s = 1$. The superscript s also indexes the corresponding recourse variables, with $\mathbf{g}^s_0 := \mathbf{g}_0$. Applying the scenario-based CVaR reformulation [7] yields the following LP.

$$(\mathcal{P}_{\mathrm{LP}}) \quad \max_{\substack{\mathbf{c}^{\mathrm{F}},\mathbf{c},\mathbf{g},\mathbf{q},\\ \mathbf{z},\boldsymbol{\tau},\boldsymbol{\xi}}} \; W(\mathbf{c}^{\mathrm{F}}, \mathbf{c}) \tag{14a}$$

$$\text{s.t.} \quad \mathbf{q}^s_t = \mathbf{g}^s_t - \mathbf{D}^s_t - \mathbf{c}^{\mathrm{F}}$$

$$-\mathbf{C}(\mathbf{c}) + \mathbf{z}^s_t, \qquad \forall t,s, \tag{14b}$$

$$\mathbf{0} \le \mathbf{z}^s_t \le \mathbf{C}(\mathbf{c}), \qquad \forall t,s, \tag{14c}$$

$$\mathbf{1}^\top \mathbf{q}^s_t = 0, \qquad \forall t,s, \tag{14d}$$

$$-\bar{\mathbf{f}} \le \mathbf{S}\mathbf{q}^s_t \le \bar{\mathbf{f}}, \qquad \forall t,s, \tag{14e}$$

$$\underline{\mathbf{g}} \le \mathbf{g}^s_t \le \bar{\mathbf{g}}, \qquad \forall t,s, \tag{14f}$$

$$-\boldsymbol{\Delta}^- \le \mathbf{g}^s_t - \mathbf{g}^s_{t-1} \le \boldsymbol{\Delta}^+, \qquad \forall t,s, \tag{14g}$$

$$\xi^s_{ik} \ge z^s_{it} - H_{ik}(\mathbf{c}_i) - \tau_{ik}, \quad \forall i,k,t,s, \tag{14h}$$

$$\tau_{ik} + \frac{1}{\epsilon_k} \sum_{s\in\mathcal{S}} p_s \xi^s_{ik} \le 0, \qquad \forall i,k, \tag{14i}$$

$$\xi^s_{ik} \ge 0, \quad \text{(5b)}, \qquad \forall i,k,s. \tag{14j}$$

To express the CVaR constraints in linear form, we introduce $\tau_{ik} \in \mathbb{R}$ and $\xi^s_{ik} \ge 0$, collected in $\boldsymbol{\tau}$ and $\boldsymbol{\xi}$. The exact linearization of the finite-scenario CVaR formulation is derived in Appendix D.

Proposition 3. *For the same finite-scenario distribution, let $(\mathbf{c}^{\mathrm{F}}, \mathbf{c}, \mathbf{g}, \mathbf{q}, \mathbf{z}, \boldsymbol{\tau}, \boldsymbol{\xi})$ be feasible for $\mathcal{P}_{\mathrm{LP}}$. Then $(\mathbf{c}^{\mathrm{F}}, \mathbf{c}, \mathbf{g}, \mathbf{q}, \mathbf{z})$ is feasible for $\mathcal{P}_{\mathrm{agg}}$. Consequently, the individual curtailments recovered by Proposition 1 form a feasible solution of $\mathcal{P}_{\mathrm{ind}}$ and satisfy* (1) *for every product.*

The proof is given in Appendix E.

Model $\mathcal{P}_{\mathrm{LP}}$ is tractable, and its feasible solutions satisfy the sampled product-level QoS limits by Proposition 3. The vector pairs $(\underline{\boldsymbol{\nu}}^{s,*}_t, \bar{\boldsymbol{\nu}}^{s,*}_t)$, $(\underline{\boldsymbol{\phi}}^{s,*}_t, \bar{\boldsymbol{\phi}}^{s,*}_t)$, and $(\underline{\boldsymbol{\psi}}^{s,*}_t, \bar{\boldsymbol{\psi}}^{s,*}_t)$ denote the nonnegative optimal dual multipliers for the lower and upper line (14e), generation (14f), and ramp bounds (14g). By the envelope theorem, the marginal values of relaxing both bounds for line $\ell \in \mathcal{J}$, generation, and ramping at bus $n \in \mathcal{N}$ are $\lambda_\ell := \sum_{t,s}(\underline{\nu}^{s,*}_{\ell t} + \bar{\nu}^{s,*}_{\ell t})$, $\lambda^{\mathrm{G}}_n := \sum_{t,s}(\underline{\phi}^{s,*}_{nt} + \bar{\phi}^{s,*}_{nt})$, and $\lambda^{\mathrm{R}}_n := \sum_{t,s}(\underline{\psi}^{s,*}_{nt} + \bar{\psi}^{s,*}_{nt})$.

IV. Numerical Study

We use the PJM 5-bus system to examine model equivalence, sensitivity to QoS parameters, and upgrade values, and the IEEE RTS 73-bus system to assess scalability.

A. PJM 5-Bus System

We first consider the PJM 5-bus system [8], with a data center at bus D. The five PJM buses are denoted by A, B, ..., E. Nominal bus loads are independently scaled by $1+\eta$, where $\eta \sim \mathcal{N}(0, 0.05^2)$ is truncated to $[-0.10, 0.10]$. For the three QoS tiers, the data center requests up to 200 MW of firm capacity and 120 MW per flexible tier. We set $\boldsymbol{\delta} = (0.01, 0.05, 0.10)$, $\boldsymbol{\epsilon} = (0.05, 0.10, 0.15)$, $w^{\mathrm{F}} = 1$, and $\mathbf{w} = (0.9, 0.8, 0.7)$. These weights express relative preferences for capacity at different reliability levels. All models use the empirical distribution with $p_s = 1/S$, and models in each comparison use the same demand samples. We use $S = 500$ for Table I and $S = 2000$ for the other experiments. We ran all experiments using Gurobi 13.0.3 (default settings) on a Windows PC with a 20-core Intel Core Ultra 7 265 processor and 32 GiB of RAM.

To assess the benefit of aggregation, we compare $\mathcal{P}_{\mathrm{ind}}$ and $\mathcal{P}_{\mathrm{agg}}$. Table I reports the same globally optimal objective value $W^\star$ for both models, as predicted by Theorem 1; their capacity awards also coincide in this test. Aggregation removes the 2500 priority binaries. The remaining 1500 binaries enforce the chance constraints.

Then, we use the duals of $\mathcal{P}_{\mathrm{LP}}$ to evaluate PJM upgrades ($S = 2000$). We compute the marginal upgrade values in Table II. Ramp limits are nonbinding in this case. Table II identifies line D–E as the most valuable marginal upgrade, followed by generation at D, C, and A.

TABLE I
COMPARISON OF THE INDIVIDUAL AND AGGREGATE MODELS ($S = 500$)

Model	$W^\star$	c^{F}	c_1	c_2	c_3	Binaries	Time (s)
$\mathcal{P}_{\mathrm{ind}}$	298.28	200.0	98.70	8.26	4.06	4000	5801.5
$\mathcal{P}_{\mathrm{agg}}$	298.28	200.0	98.70	8.26	4.06	1500	1.56

TABLE II
MARGINAL UPGRADE VALUES FROM $\mathcal{P}_{\mathrm{LP}}$ ($S = 2000$)

Facility	Line/generation capacity limits (MW)	Marginal upgrade value λ
Line D–E	240	1.92
Generation at D	200	0.92
Generation at C	520	0.62
Generation at A	210	0.21

Next, we study the two QoS parameters separately to see how each affects flexible capacity. With one flexible tier and no firm capacity, we maximize the flexible award, and the flexible request limit is relaxed. Fig. 2 compares the aggregate and CVaR-based LP results. Both admit more capacity when either δ or ϵ increases. In the aggregate model, raising δ from 0.01 to 0.20 increases the award from 306.4 to 379.1 MW (24%), while raising ϵ from 0.01 to 0.20 increases it by 11%. The LP award is approximately 96.8% of the aggregate award.

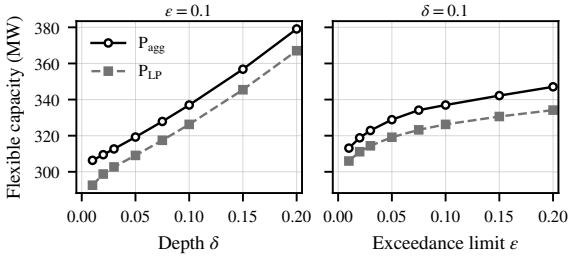

Fig. 2. Effect of curtailment depth δ (left, $\epsilon = 0.1$) and exceedance limit ϵ (right, $\delta = 0.1$) on single-tier flexible capacity ($S = 2000$, $c^{\mathrm{F}} = 0$).

B. IEEE RTS 73-Bus System

We also test the method on the PGLib-OPF IEEE RTS 73-bus system [9], using 24-h trajectories [10]. We apply multiplicative Gaussian perturbations at the daily level ($\sigma = 2\%$) and independently across bus-hours ($\sigma = 3\%$), truncating both at $\pm 2\sigma$. Three data centers, at buses 103, 214, and 319, each request up to 400 MW of firm capacity and 200 MW per flexible tier, with the same QoS parameters and weights as in the PJM case. Global optimality is not proven for either $\mathcal{P}_{\mathrm{ind}}$ or $\mathcal{P}_{\mathrm{agg}}$ within two hours. We therefore report the results of $\mathcal{P}_{\mathrm{LP}}$, which is solved in 245 s.

TABLE III
RTS-73 CAPACITY AWARDS (MW) AND MAXIMUM IN-SAMPLE DEPTH-EXCEEDANCE RATES ($S = 2000$)

	c^{F}	Tier 1	Tier 2	Tier 3
Bus 103	400.0	0.0	0.0	55.8
Bus 214	400.0	0.0	0.0	128.8
Bus 319	400.0	144.9	154.0	200.0
Total	1200.0	144.9	154.0	384.6
Max. in-sample (%)	–	0.15	0.70	3.75

For each active product, we recover individual curtailments using (11) and compute the fraction of the 2000 trajectories in which its allowed depth is exceeded. Table III reports the maximum of these fractions across active products in each tier. All rates remain below their limits, consistent with the in-sample guarantee of Proposition 3. All three firm requests are awarded in full, and the total admitted capacity is 1883.5 MW. This is 45.0% above the 1298.8 MW firm-only benchmark, computed on the same demand trajectories with relaxed firm-request upper bounds.

V. CONCLUSION

This paper studies large-load interconnection with firm capacity and multiple flexible capacity tiers under a two-layer curtailment priority rule. We show that the individual and aggregate quantification models have the same feasible set of capacity decisions. This removes the priority binaries without changing the optimal value, thus supporting the efficient quantification of hosting capacity. The CVaR approximation further yields an LP that satisfies the sampled individual QoS limits and provides dual information for screening possible grid upgrades.

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APPENDIX A
PROOF OF PROPOSITION 1

Proof. Fix $z \in [0, C_i]$ and define x_{im} by (10). Set $\beta_{i0} = 1$, $\beta_{i,2K} = 0$, and $\beta_{im} := \mathbf{1}\{z \geq \Gamma_{im}\}$ for $0 < m < 2K$. Here, $\mathbf{1}\{\cdot\}$ denotes the indicator function. Obtain y_{ik} from (4e). Since the intervals $[\Gamma_{i,m-1}, \Gamma_{im}]$ cover $[0, C_i]$, (10) allocates z across the tranches in order, giving $0 \leq x_{im} \leq d_{im}$ and $\sum_m x_{im} = z$. We verify that this allocation satisfies the priority constraints. If $\beta_{i,m-1} = 0$, then $z < \Gamma_{i,m-1}$ and $x_{im} = 0$; otherwise, $x_{im} \leq d_{im} \leq M$. Thus, (4b) holds. Similarly, if $\beta_{im} = 1$, then $z \geq \Gamma_{im}$ and $x_{im} = d_{im}$; otherwise, the lower bound in (4c) is nonpositive. Since Γ_{im} is nondecreasing in m, the binary variables satisfy (4d) and (4f). Finally, (4e) assigns each tranche to one product, so $\sum_k y_{ik} = \sum_m x_{im} = z$. Hence all constraints in (4) are satisfied. Conversely, consider any feasible $(\mathbf{x}, \mathbf{y}, \boldsymbol{\beta})$ with total curtailment z. By (4d) and (4f), the binary variables form a nonincreasing sequence from 1 to 0. Let $m_0 := \min\{m : \beta_{im} = 0\}$. For $m < m_0$, constraint (4c) forces $x_{im} = d_{im}$, while for $m > m_0$, constraint (4b) forces $x_{im} = 0$. Thus, only tranche m_0 can be partially curtailed. Since $\sum_m x_{im} = z$, we must have

$$x_{i,m_0} = z - \Gamma_{i,m_0-1}, \qquad \Gamma_{i,m_0-1} \leq z \leq \Gamma_{i,m_0}.$$

Therefore, all tranche curtailments are uniquely determined by z and are given by (10).

Each product k consists of tranches $K - k + 1$ and $2K - k + 1$, with sizes a_{ik} and b_{ik} and starting points A_{ik} and H_{ik}, respectively. Applying (10) to these two tranches and summing their curtailments gives (11). □

APPENDIX B
PROOF OF PROPOSITION 2

Proof. For an active product, $b_{ik} = (1 - \delta_k)c_{ik} > 0$. By definition, H_{ik} is the aggregate curtailment level at which the exceedance tranche of product k starts. Moreover, from (9),

$$H_{ik} - A_{ik} - a_{ik} = \sum_{r<k} a_{ir} + \sum_{r>k} b_{ir} \geq 0,$$

so the within-depth tranche is fully curtailed before the exceedance tranche can be used.

Fix $z \in [0, C_i]$. If $z \leq H_{ik}$, the exceedance tranche receives no curtailment, and (11) gives $f_{ik}(z; \mathbf{c}_i) \leq a_{ik} = \delta_k c_{ik}$. If $z > H_{ik}$, the within-depth tranche is fully curtailed and the exceedance tranche receives positive curtailment, so $f_{ik}(z; \mathbf{c}_i) > \delta_k c_{ik}$. Hence, $f_{ik}(z; \mathbf{c}_i) > \delta_k c_{ik}$ if and only if $z > H_{ik}$.

By Proposition 1, $y_{ikt} = f_{ik}(z_{it}; \mathbf{c}_i)$ in every period. Therefore, product k exceeds its allowed depth during the planning horizon if and only if the aggregate curtailment exceeds H_{ik} in at least one period:

$$\max_{t \in \mathcal{T}} y_{ikt} > \delta_k c_{ik} \iff \exists t \in \mathcal{T} : \; z_{it} > H_{ik} \iff Z_i > H_{ik}.$$

This establishes (12). Since the two events coincide in every realization, their probabilities are equal, which proves the equivalence of the two chance constraints. □

APPENDIX C
PROOF OF THEOREM 1

Proof. From $\mathcal{P}_{\mathrm{ind}}$ to $\mathcal{P}_{\mathrm{agg}}$. Take a feasible solution of $\mathcal{P}_{\mathrm{ind}}$ and set $z_{it} := \sum_k y_{ikt}$. By (4a) and (4e), $0 \leq y_{ikt} \leq a_{ik} + b_{ik} = c_{ik}$, so (7b) holds. Substituting (6) into (5c) gives (7a) with the same nodal injections, while (5d)–(5g) remain unchanged. For each active product, Proposition 2 shows that (5h) implies (13c). For an inactive product (i, k), suppose there is an active tier with index smaller than k and let $h := \max\{r < k : c_{ir} > 0\}$. All tiers r with $h < r \leq k$ are inactive, so $H_{ik} = H_{ih}$ by (9). Since $\epsilon_h \leq \epsilon_k$ by (2), we have $\mathbb{P}(Z_i > H_{ik}) = \mathbb{P}(Z_i > H_{ih}) \leq \epsilon_h \leq \epsilon_k$. If no such h exists, then $c_{ir} = 0$ for all $r \leq k$, so $H_{ik} = C_i \geq Z_i$ and the exceedance event is empty.

From $\mathcal{P}_{\mathrm{agg}}$ to $\mathcal{P}_{\mathrm{ind}}$. Take a feasible solution of $\mathcal{P}_{\mathrm{agg}}$. For each candidate bus, period, and realization, (7b) gives $z_{it} \in [0, C_i]$. By Proposition 1, set $y_{ikt} := f_{ik}(z_{it}; \mathbf{c}_i)$ and obtain x_{imt} from (10). Set $\beta_{i0t} = 1$, $\beta_{i,2K,t} = 0$, and $\beta_{imt} := \mathbf{1}\{z_{it} \geq \Gamma_{im}\}$ for $0 < m < 2K$. These variables satisfy (4) with $\sum_k y_{ikt} = z_{it}$. The recovered $\mathbf{x}$ and $\mathbf{y}$ are continuous functions of z_{it}, and the binary variables are indicator functions of z_{it}, so these are valid recourse decisions. Since $\sum_k y_{ikt} = z_{it}$, (5c) gives the same nodal injections, and (5d)–(5g) remain satisfied. For each active product, Proposition 2 turns (13c) into (5h). For an inactive product, $y_{ikt} = 0$, so its exceedance event is empty. The recovered solution is therefore feasible for $\mathcal{P}_{\mathrm{ind}}$.

Both mappings preserve $(\mathbf{c}^{\mathrm{F}}, \mathbf{c})$, so the two problems have the same feasible set of capacity decisions. Since their common objective W depends only on these capacities, their optimal values are also equal, and an optimal solution of either problem is mapped to an optimal solution of the other. □

APPENDIX D
DERIVATION OF THE CVAR-BASED LINEAR PROGRAM

The loss of product (i, k) in trajectory s is

$$L^s_{ik} := \max_{t \in \mathcal{T}} z^s_{it} - H_{ik}(\mathbf{c}_i). \tag{15}$$

The minimization representation of CVaR [11] gives

$$\mathrm{CVaR}_{1-\epsilon_k}(L_{ik}) = \min_{\tau_{ik} \in \mathbb{R}} \left\{ \tau_{ik} + \frac{1}{\epsilon_k} \sum_{s \in \mathcal{S}} p_s [L^s_{ik} - \tau_{ik}]_+ \right\}. \tag{16}$$

For a finite scenario set and $0 < \epsilon_k < 1$, this minimum is attained. Hence, $\mathrm{CVaR}_{1-\epsilon_k}(L_{ik}) \le 0$ holds if and only if there exist $\tau_{ik} \in \mathbb{R}$ and auxiliary variables ξ_{ik}^s such that

$$\tau_{ik} + \frac{1}{\epsilon_k} \sum_{s \in \mathcal{S}} p_s \xi_{ik}^s \le 0, \tag{17a}$$

$$\xi_{ik}^s \ge [L_{ik}^s - \tau_{ik}]_+, \quad \forall s \in \mathcal{S}. \tag{17b}$$

Indeed, a feasible CVaR constraint admits a minimizing τ_{ik} and the choice $\xi_{ik}^s := [L_{ik}^s - \tau_{ik}]_+$. Conversely, any variables satisfying (17) provide an upper bound on (16) that is at most zero.

Using (15), the excess loss can be written as

$$[L_{ik}^s - \tau_{ik}]_+ = \max\Big\{0, \max_{t \in \mathcal{T}} \big\{ z_{it}^s - H_{ik}(\mathbf{c}_i) - \tau_{ik} \big\}\Big\}. \tag{18}$$

Therefore, (17b) is equivalent to

$$\begin{aligned} &\xi_{ik}^s \ge 0, \\ &\xi_{ik}^s \ge z_{it}^s - H_{ik}(\mathbf{c}_i) - \tau_{ik}, \quad \forall t \in \mathcal{T}. \end{aligned} \tag{19}$$

Thus, one variable ξ_{ik}^s per product and trajectory accounts for the maximum over all periods. Imposing (17a) and (19) for every (i,k) and s yields (14h)–(14i) and $\xi_{ik}^s \ge 0$.

Retaining the capacity bounds (5b) and imposing the network constraints for each trajectory gives (14b)–(14g). Since the objective is linear and $\mathbf{C}(\mathbf{c})$ and $H_{ik}(\mathbf{c}_i)$ are affine in the capacity decisions, the resulting formulation is an LP. Introducing $\boldsymbol{\tau}$ and $\boldsymbol{\xi}$ preserves the objective and feasible decisions in the original variables, so $\mathcal{P}_{\mathrm{LP}}$ is equivalent to the finite-scenario CVaR formulation.

Appendix E
Proof of Proposition 3

Proof. Fix (i,k), write $\epsilon := \epsilon_k$ and $\tau := \tau_{ik}$, and let $L^s := \max_t\{z_{it}^s - H_{ik}(\mathbf{c}_i)\}$. By (14h) and (14j), $\xi_{ik}^s \ge [L^s - \tau]_+$. Thus, (14i) gives

$$\tau + \frac{1}{\epsilon} \sum_{s \in \mathcal{S}} p_s [L^s - \tau]_+ \le 0,$$

which implies $\tau \le 0$. If $\tau = 0$, then $L^s \le 0$ for every s because $p_s > 0$. If $\tau < 0$, then $[L^s - \tau]_+ \ge (-\tau)\mathbf{1}\{L^s > 0\}$, and therefore

$$\begin{aligned} \mathbb{P}\big(Z_i > H_{ik}(\mathbf{c}_i)\big) &= \sum_{s \in \mathcal{S}} p_s \mathbf{1}\{L^s > 0\} \\ &\le \frac{1}{-\tau} \sum_{s \in \mathcal{S}} p_s [L^s - \tau]_+ \le \epsilon. \end{aligned}$$

In both cases, (13c) holds. The remaining constraints of $\mathcal{P}_{\mathrm{agg}}$ are also satisfied by $\mathcal{P}_{\mathrm{LP}}$, so the solution is feasible for $\mathcal{P}_{\mathrm{agg}}$. Theorem 1 then gives feasibility for $\mathcal{P}_{\mathrm{ind}}$ and the product-level reliability guarantee. □